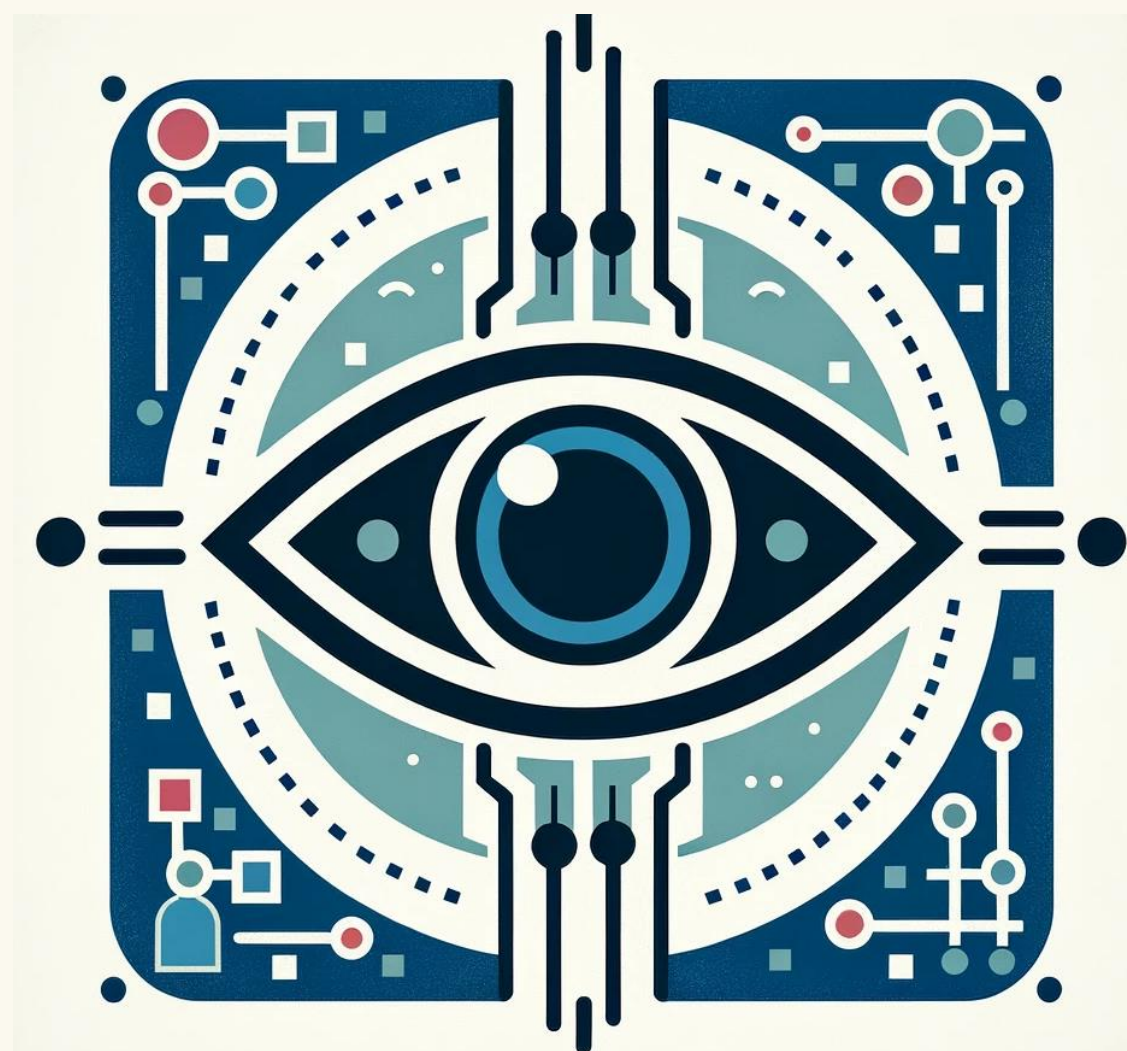




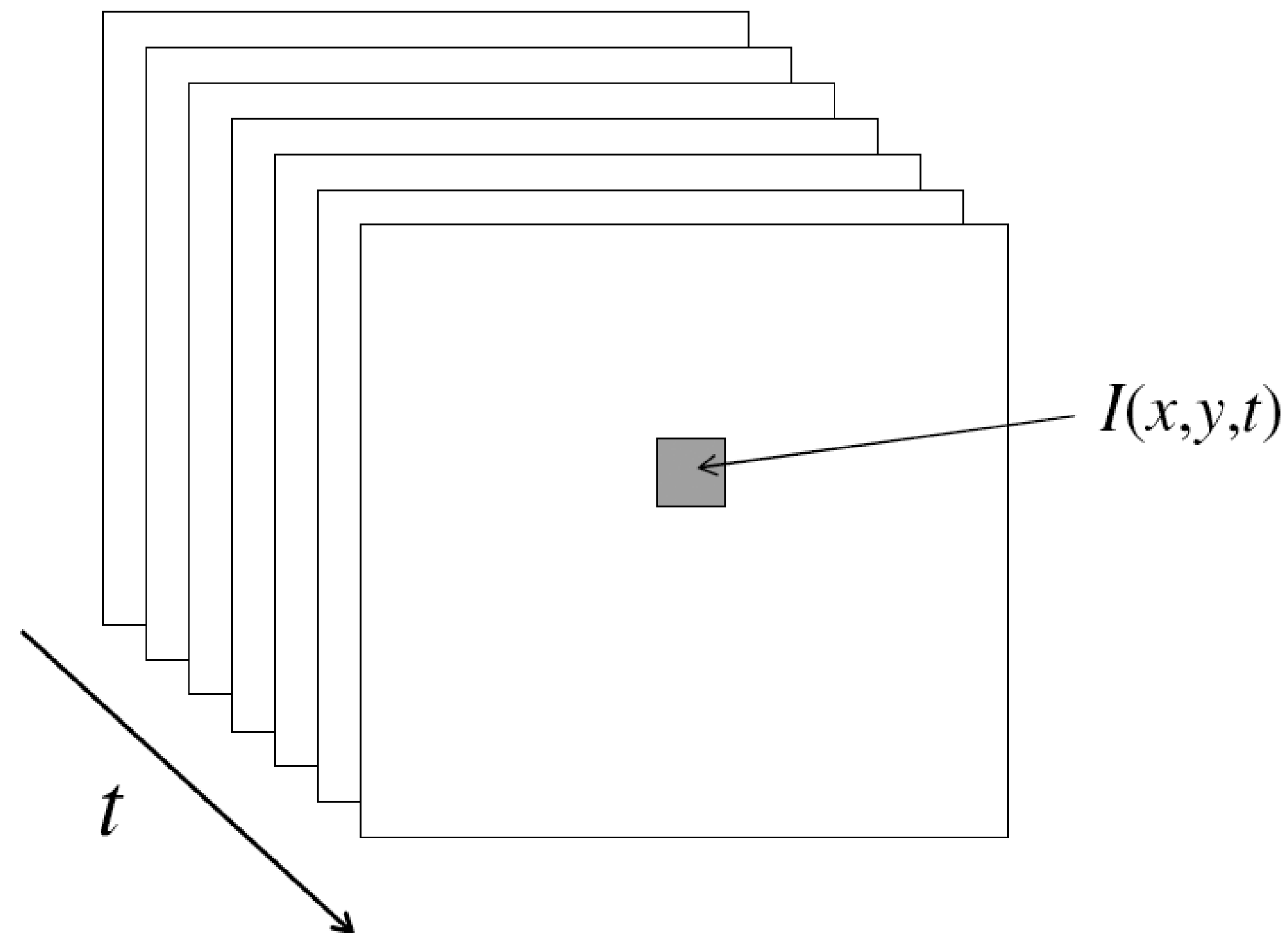
COMPUTER VISION

Motion

Image Sequences

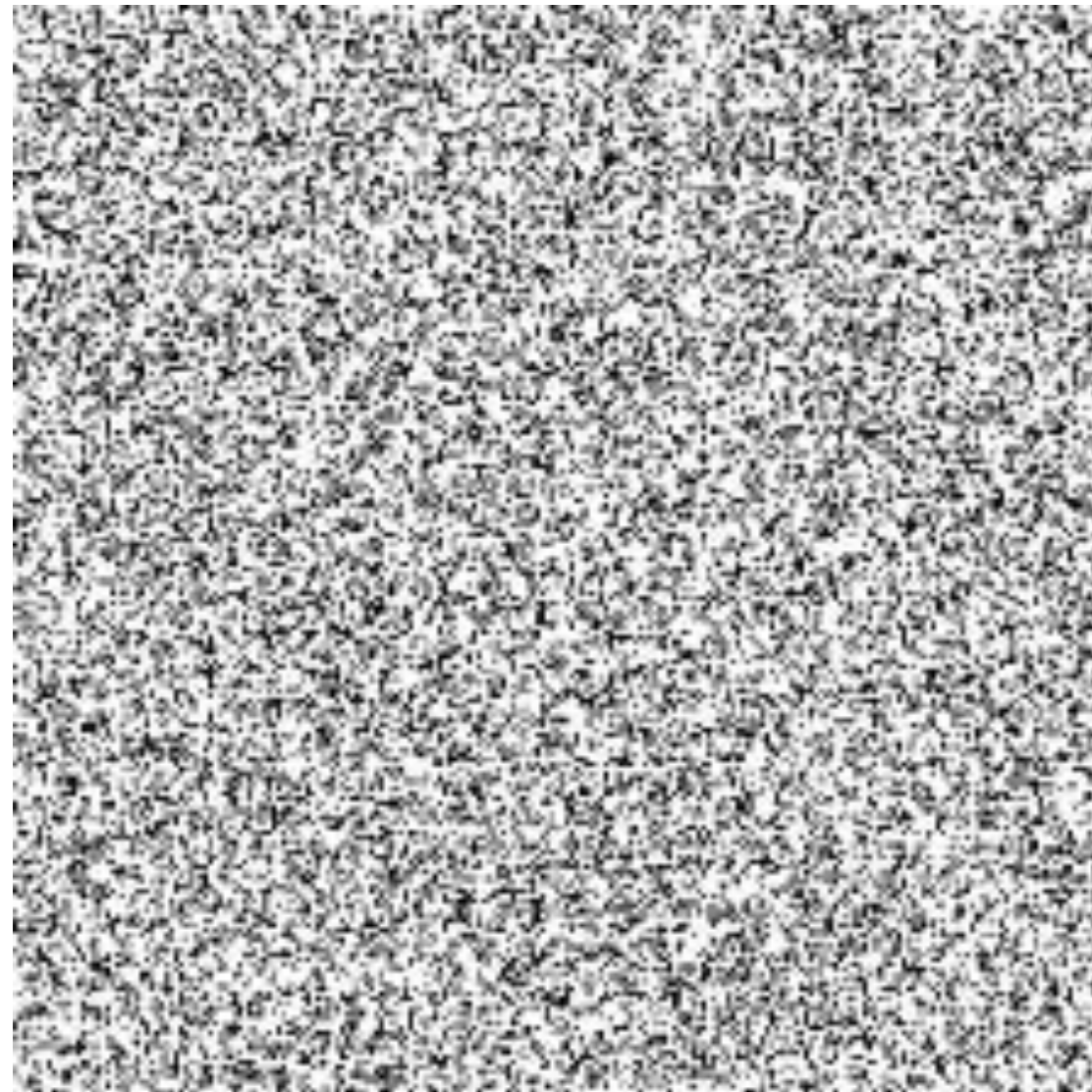
A **video** is a **sequence of images** (frames) captured over time

Now our image data is a function of **space** (x, y) and **time** (t)



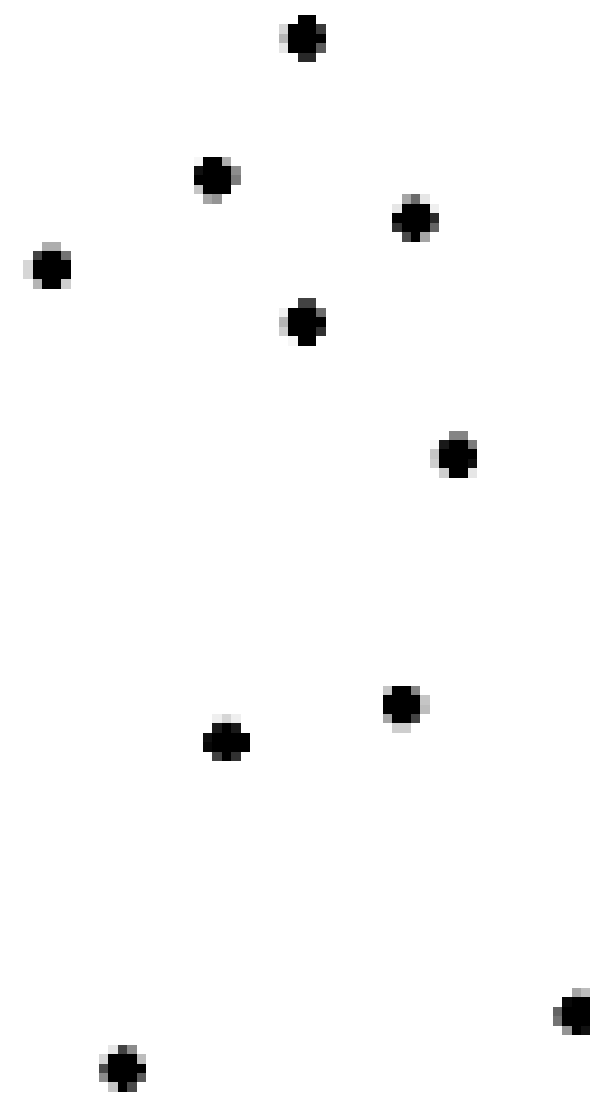
Motion and perceptual organization

Sometimes, motion is the only cue



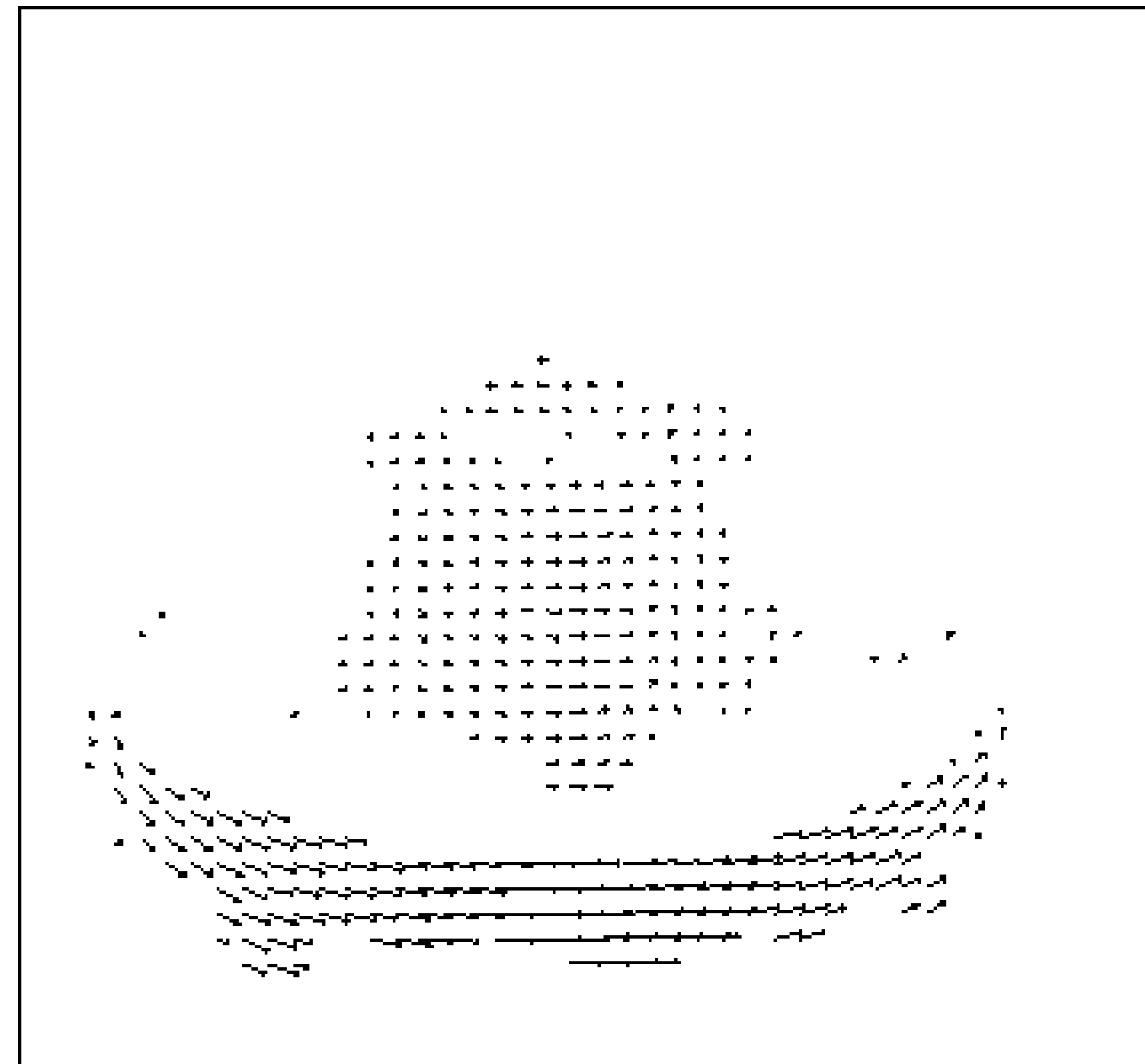
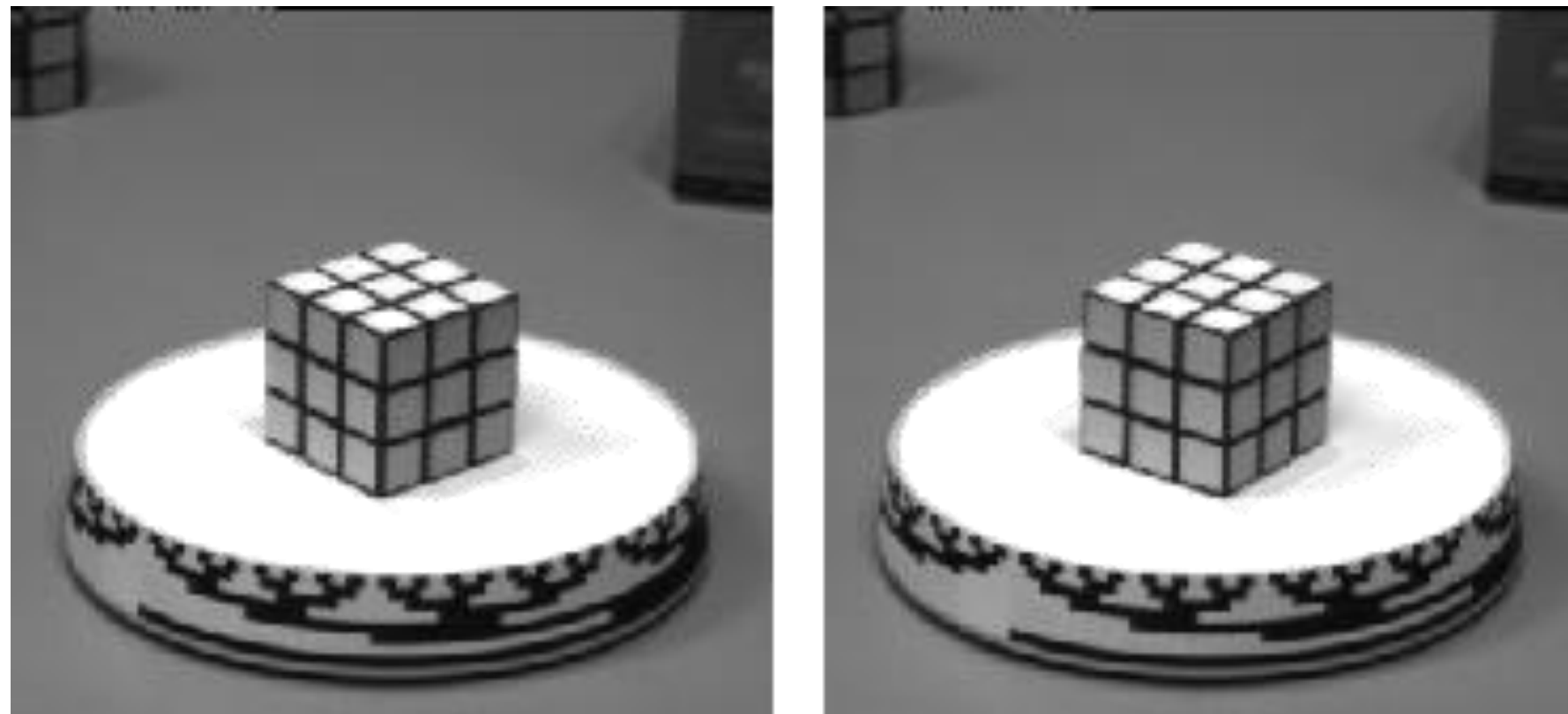
Motion and perceptual organization

Even “impoverished” motion data can evoke a strong percept

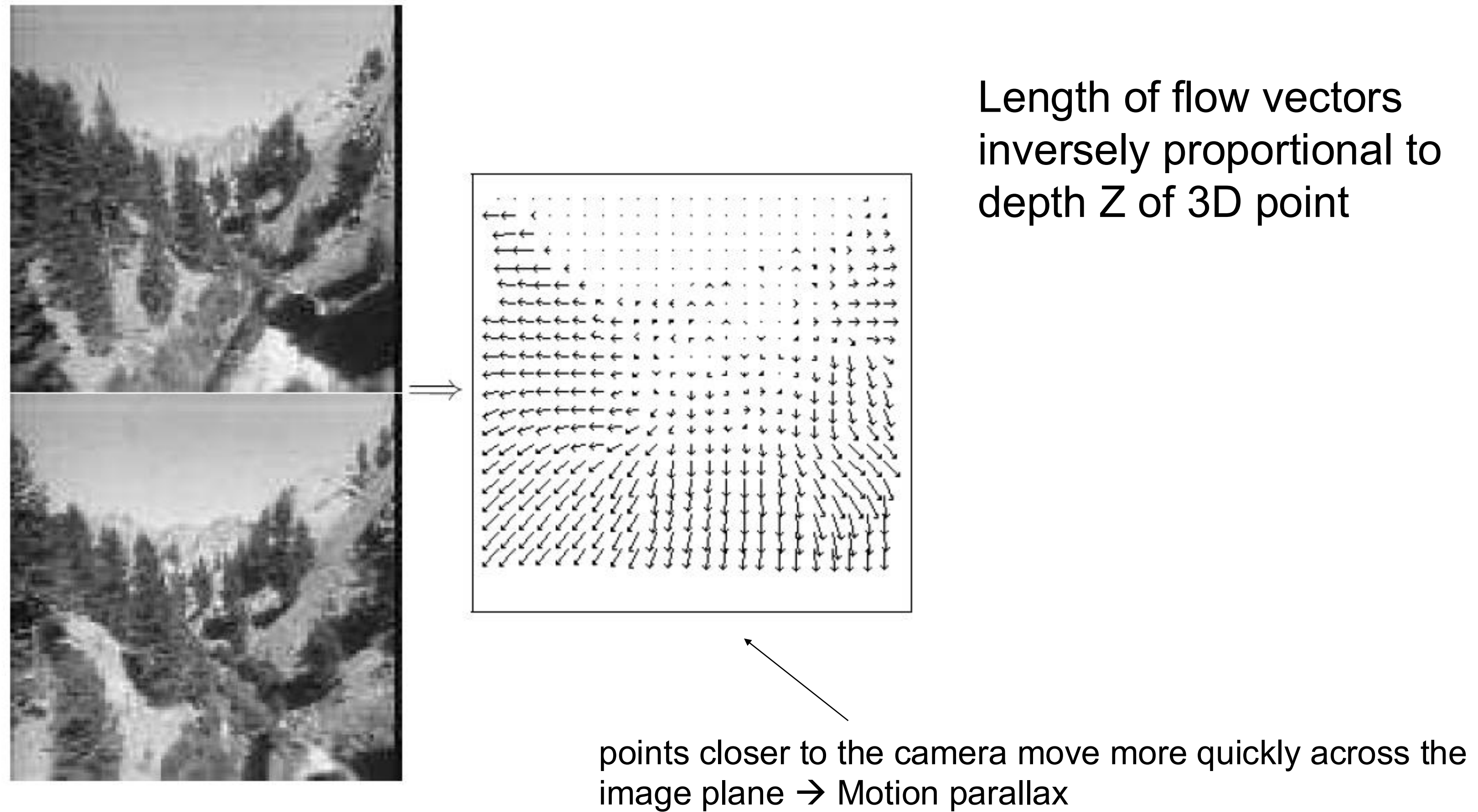


Motion field

The motion field is the projection of the 3D scene motion into the image

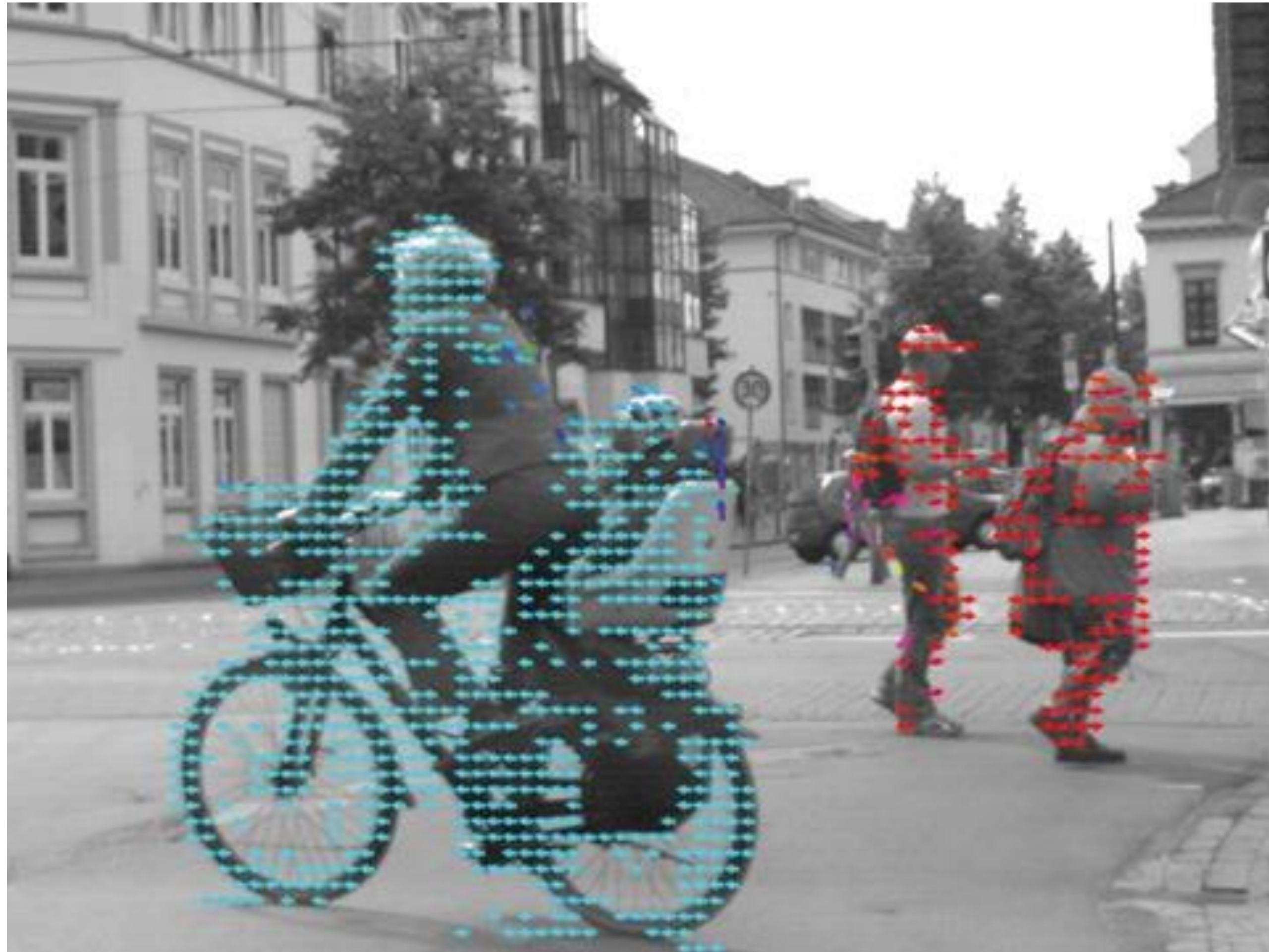


Motion field + camera motion



Optical flow

Vector field function of the spatio-temporal **image brightness variations**



Optical flow

Definition: optical flow is the *apparent* motion of brightness patterns in the image

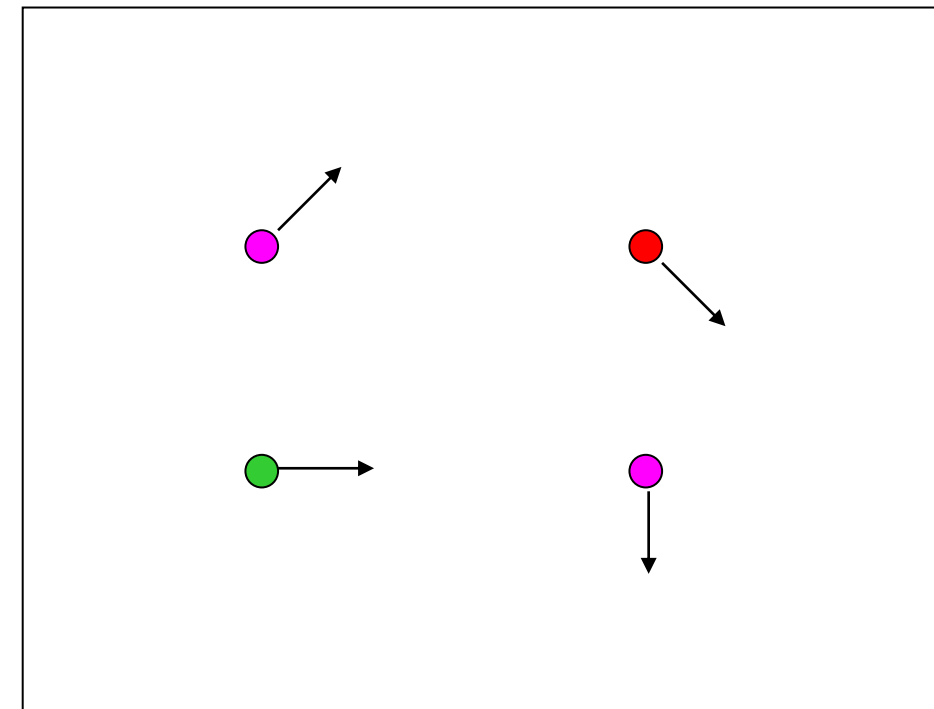
Ideally, optical flow would be the same as the motion field

However, the apparent motion can be caused by lighting changes without any actual motion

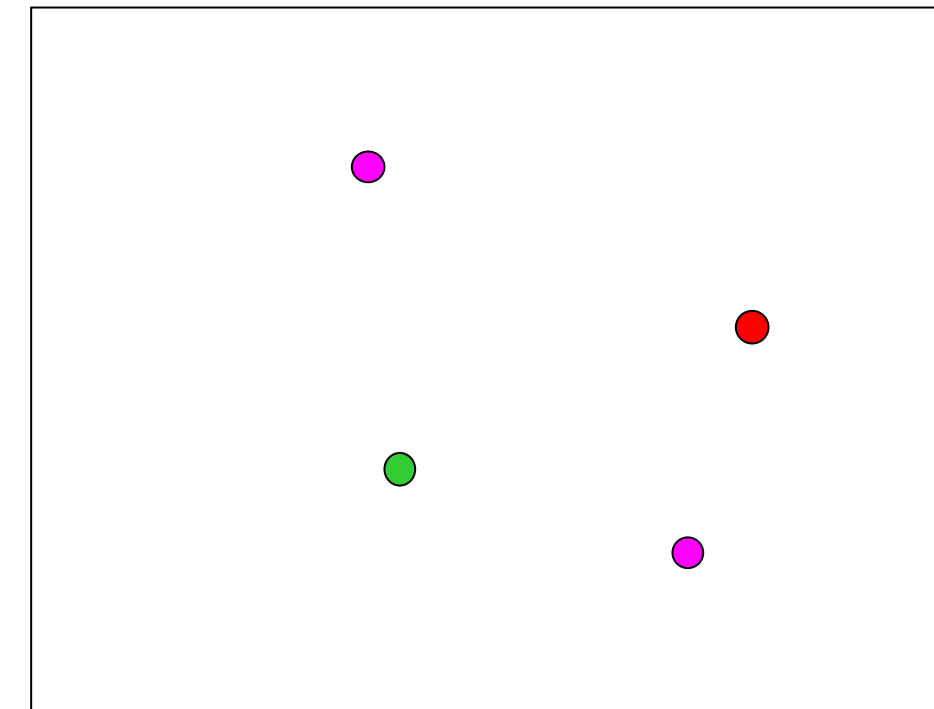
Estimating the Optical flow

- Let's start by considering each pixel separately
- How do we estimate pixel motion?
 - Feature matching → Solved?
 - Can we use what we have seen before?
 - Not really... although it works well for cases of matching, detection, recognition where we only need a global alignment estimation
 - Sparse
 - Feature alignment not exact
 - Low accuracy

Feature tracking



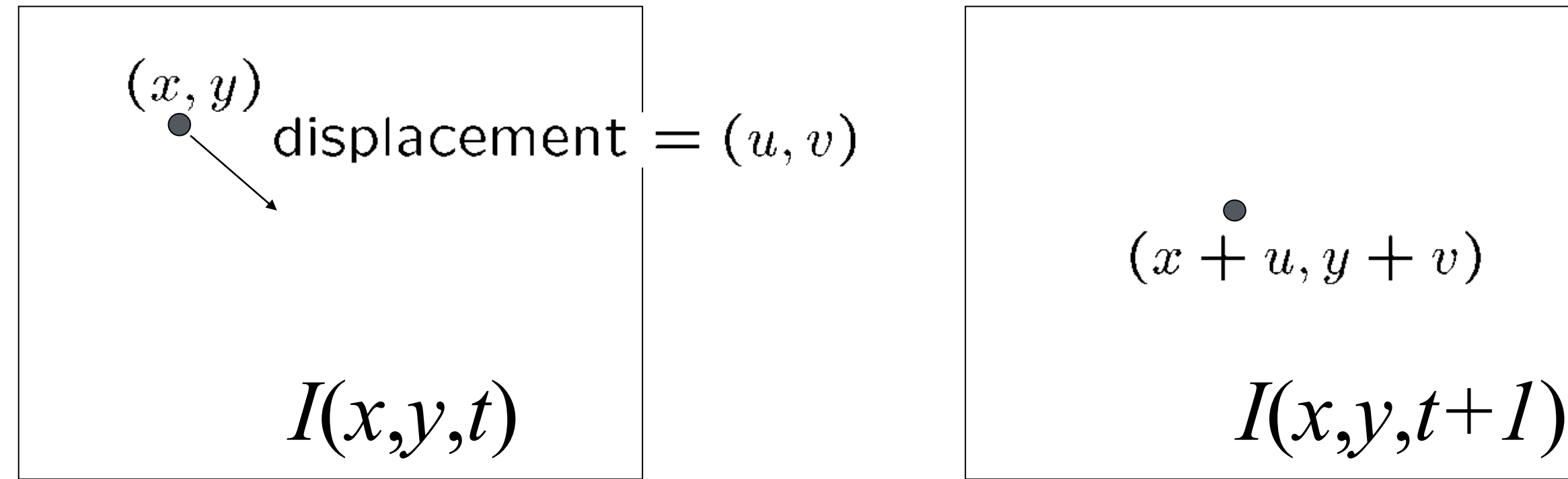
$I(x,y,t)$



$I(x,y,t+1)$

- Given two subsequent frames, estimate the point translation
- Key assumptions of *Lucas-Kanade Tracker*
 - Brightness constancy: projection of the same point looks the same in every frame
 - Small motion: points do not move very far
 - Spatial coherence: points move like their neighbors

The brightness constancy constraint



Brightness Constancy Equation:

$$I(x, y, t) = I(x + u, y + v, t + 1)$$

Small motion (one pixel):

Take first-order Taylor approximation and rearrange expression

$$I_x u + I_y v + I_t = 0$$

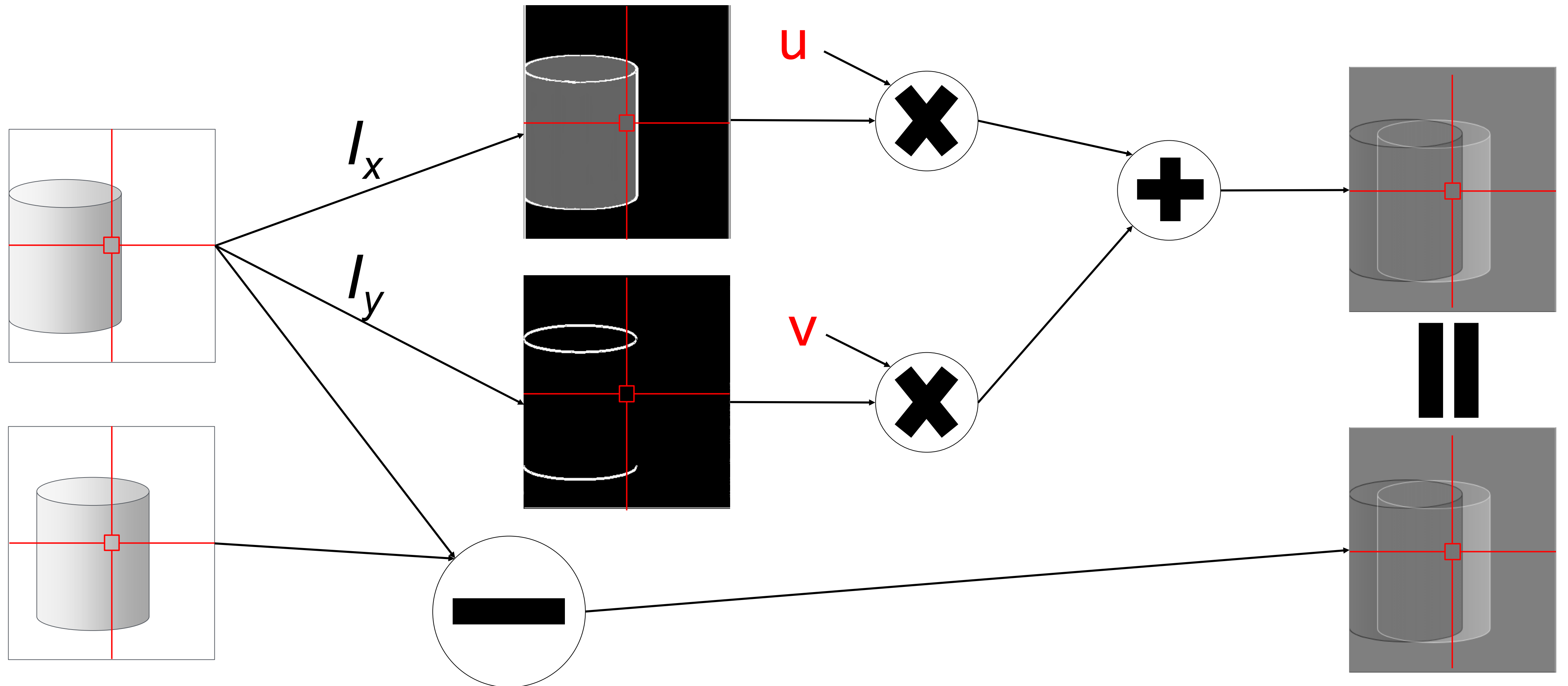
$$\nabla I \cdot [u \ v]^T + I_t = 0$$

Image derivatives Difference over frames

One equation, two unknowns (u, v)

The brightness constancy constraint

$$I_x * u + I_y * v = I_t[x,y] - I_{t+\Delta t}[x,y]$$



Solving the ambiguity

- How to get more equations for a pixel?
- Spatial coherence constraint
 - Assume the pixel's neighbors have the same (u,v)
 - If we use a 5x5 window, that gives us 25 equations per pixel

$$\begin{array}{ccc}
 A & d = & b \\
 25 \times 2 & 2 \times 1 & 25 \times 1
 \end{array}
 \begin{bmatrix} I_x(\mathbf{p}_1) & I_y(\mathbf{p}_1) \\ I_x(\mathbf{p}_2) & I_y(\mathbf{p}_2) \\ \vdots & \vdots \\ I_x(\mathbf{p}_{25}) & I_y(\mathbf{p}_{25}) \end{bmatrix}
 \begin{bmatrix} u \\ v \end{bmatrix} = - \begin{bmatrix} I_t(\mathbf{p}_1) \\ I_t(\mathbf{p}_2) \\ \vdots \\ I_t(\mathbf{p}_{25}) \end{bmatrix}$$

Overconstrained linear system

Least squares solution

$$\begin{bmatrix} \sum I_x I_x & \sum I_x I_y \\ \sum I_x I_y & \sum I_y I_y \end{bmatrix}
 \begin{bmatrix} u \\ v \end{bmatrix} = - \begin{bmatrix} \sum I_x I_t \\ \sum I_y I_t \end{bmatrix}$$

The summations are over all pixels in the K x K window

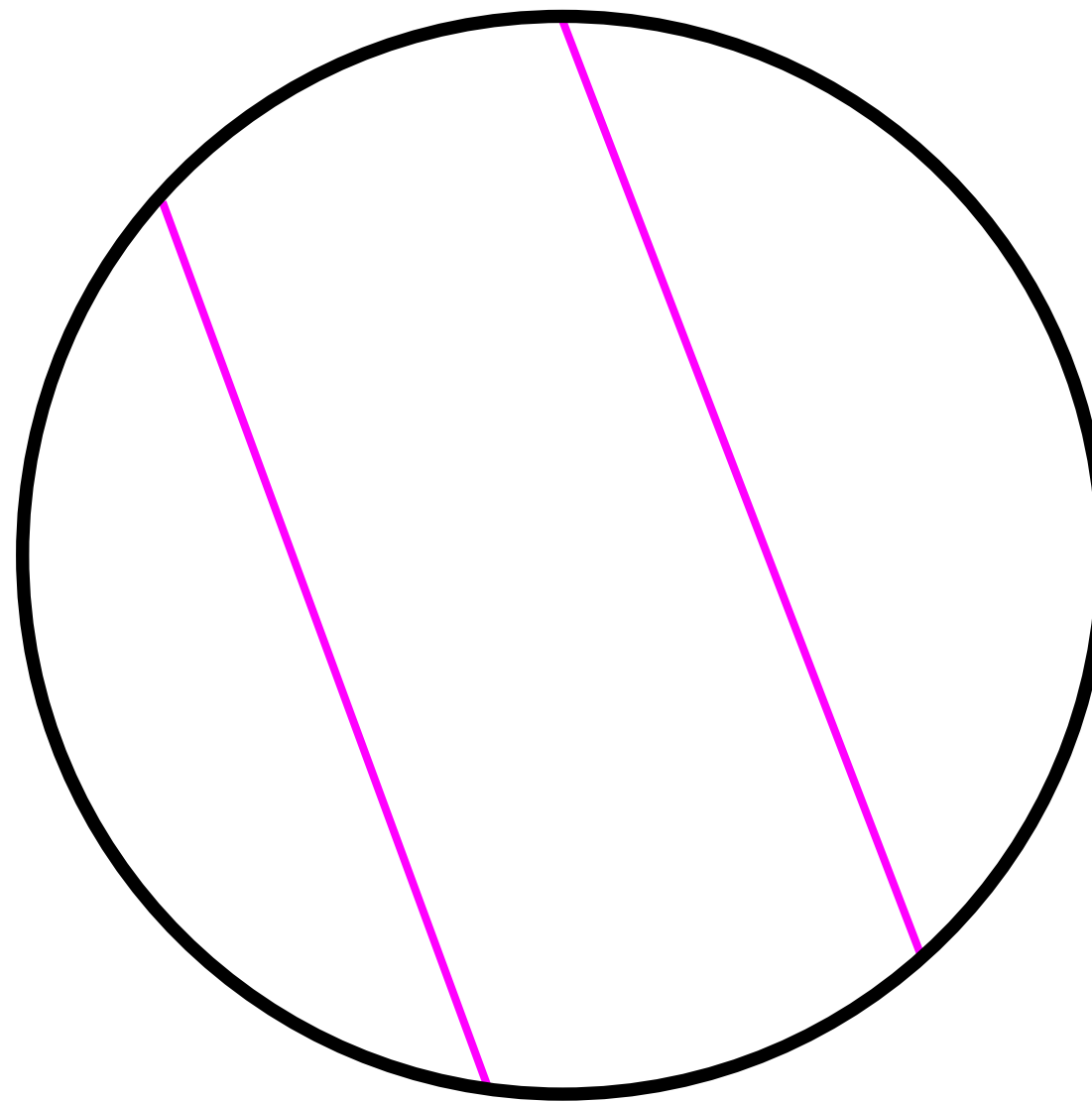
$$\begin{array}{ccc}
 A^T A & & A^T b
 \end{array}$$

Lucas-Kanade Optical Flow

When does LK fail?

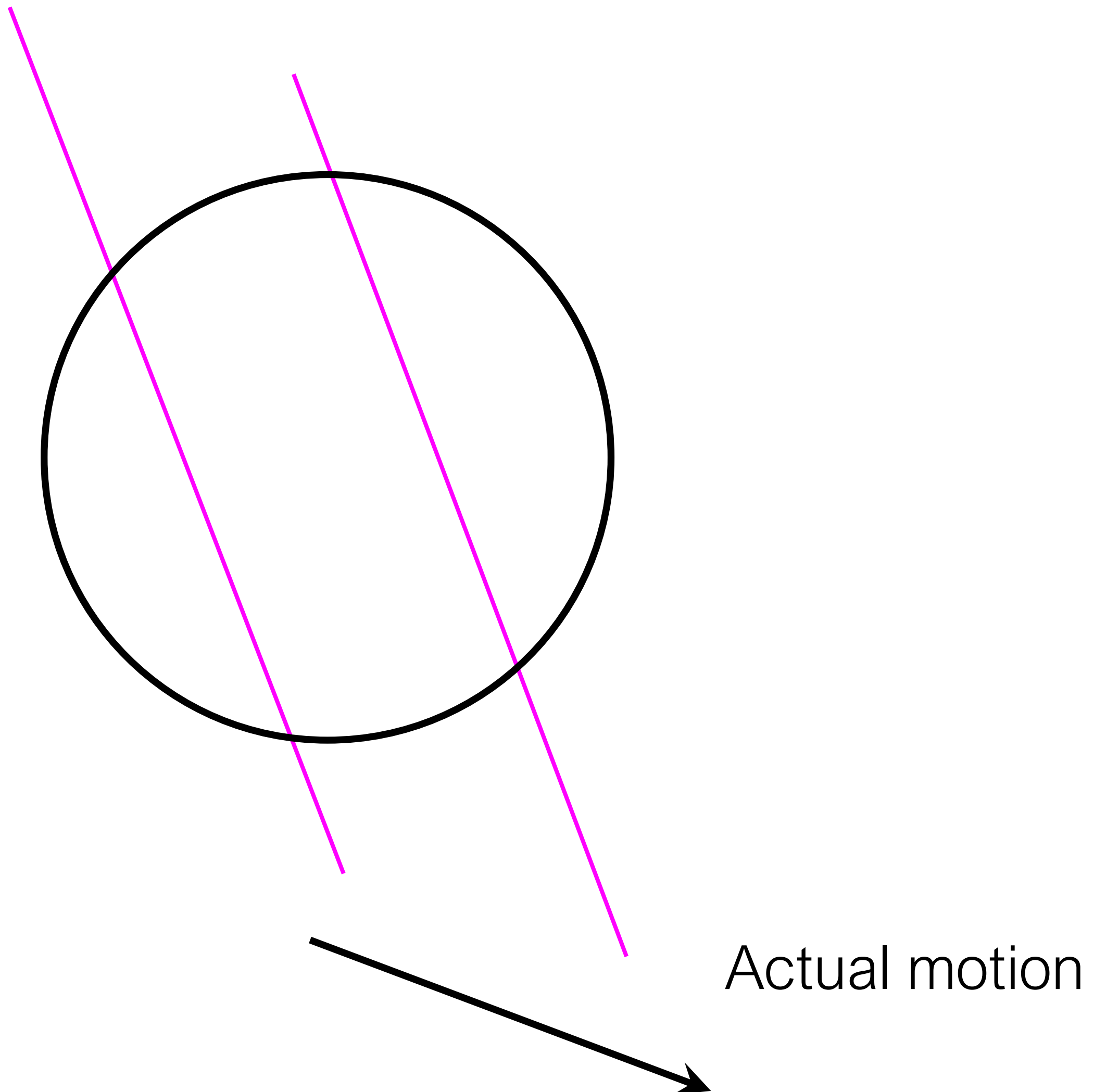
- Lighting changes
- Large movement
- Specularities
- No good features
- Aperture Problem

The aperture problem

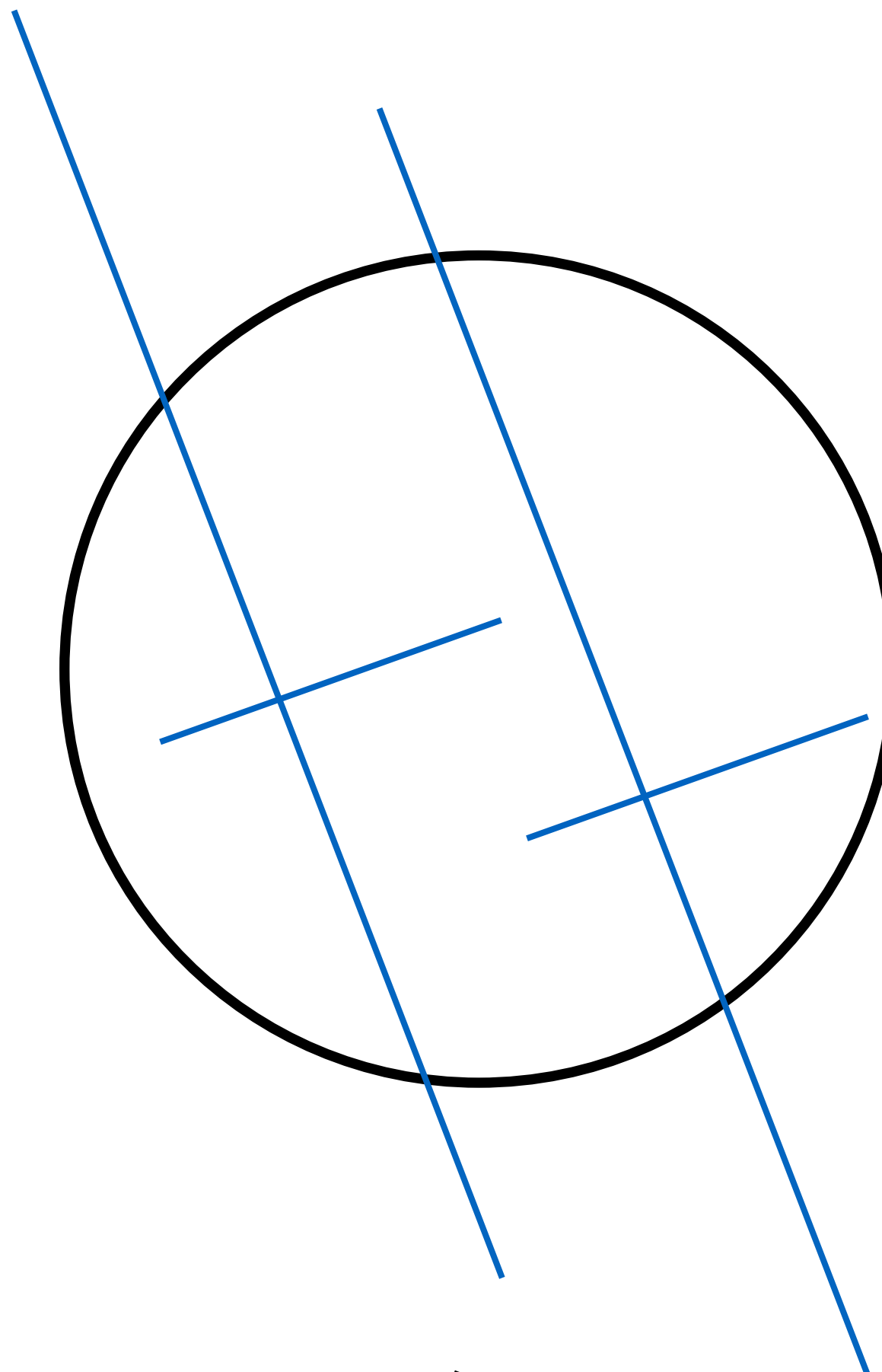


Perceived motion

The aperture problem

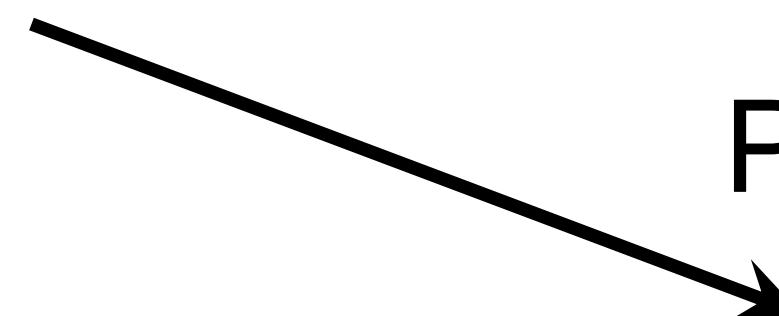
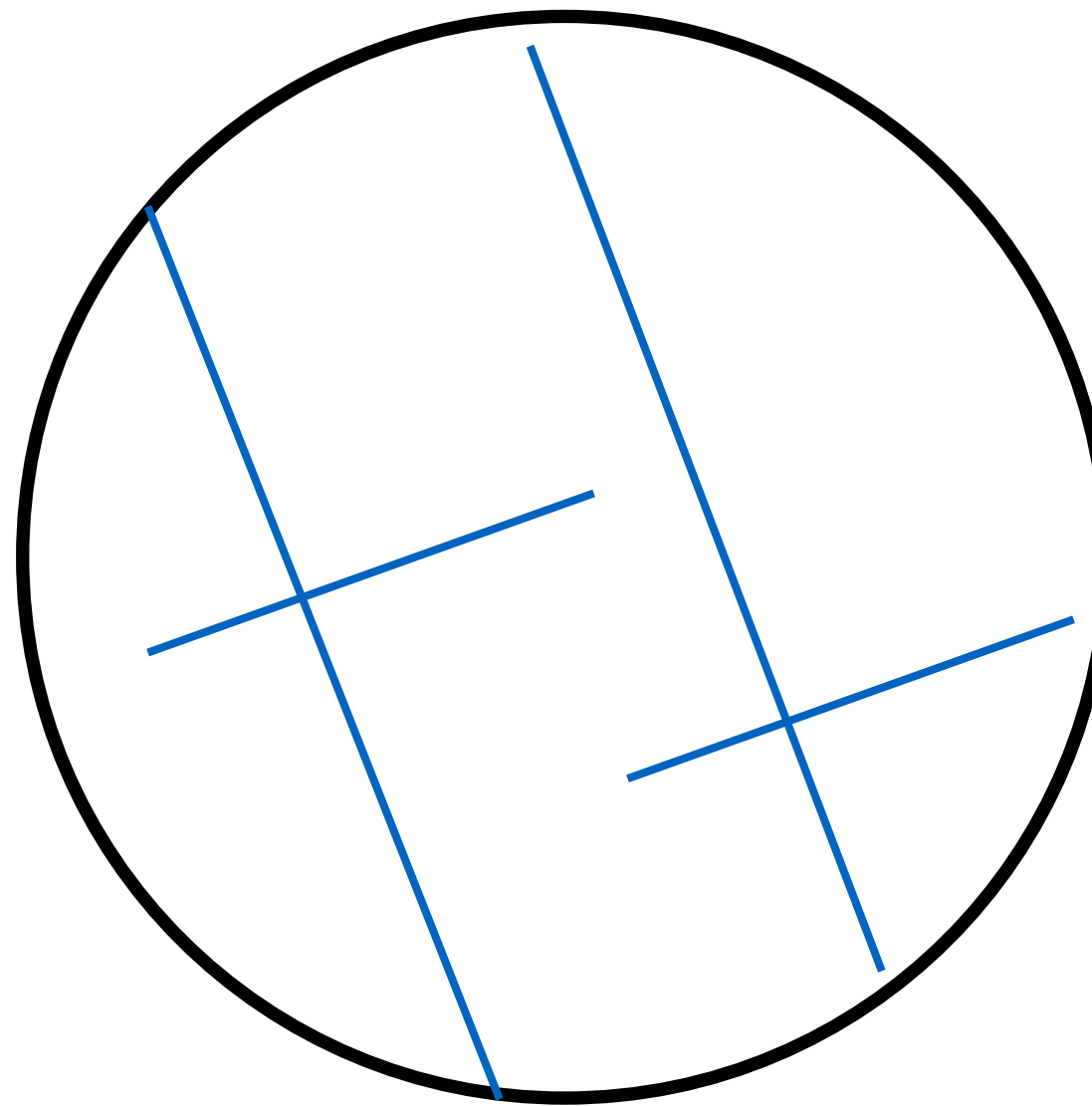


The aperture problem resolved



Actual motion

The aperture problem resolved



Perceived motion

Conditions for solvability

Optimal (u, v) satisfies Lucas-Kanade equation

$$\begin{bmatrix} \sum I_x I_x & \sum I_x I_y \\ \sum I_x I_y & \sum I_y I_y \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = - \begin{bmatrix} \sum I_x I_t \\ \sum I_y I_t \end{bmatrix}$$

$A^T A$ $A^T b$

When is this solvable? I.e., what are good points to track?

- $A^T A$ should be invertible
- $A^T A$ should not be too small due to noise
 - eigenvalues λ_1 and λ_2 of $A^T A$ should not be too small
- $A^T A$ should be well-conditioned
 - λ_1 / λ_2 should not be too large (λ_1 = larger eigenvalue)

Does this remind you of anything?

Criteria for Harris corner detector

Improvements to Lucas-Kanade Optical Flow

- Revisiting the small motion assumption:
 - What if the motion is large (larger than a pixel)?
 - Not-linear: Iterative refinement – estimate, propagate motion, repeat until convergence
 - Local minima: coarse-to-fine estimation

Image Pyramids

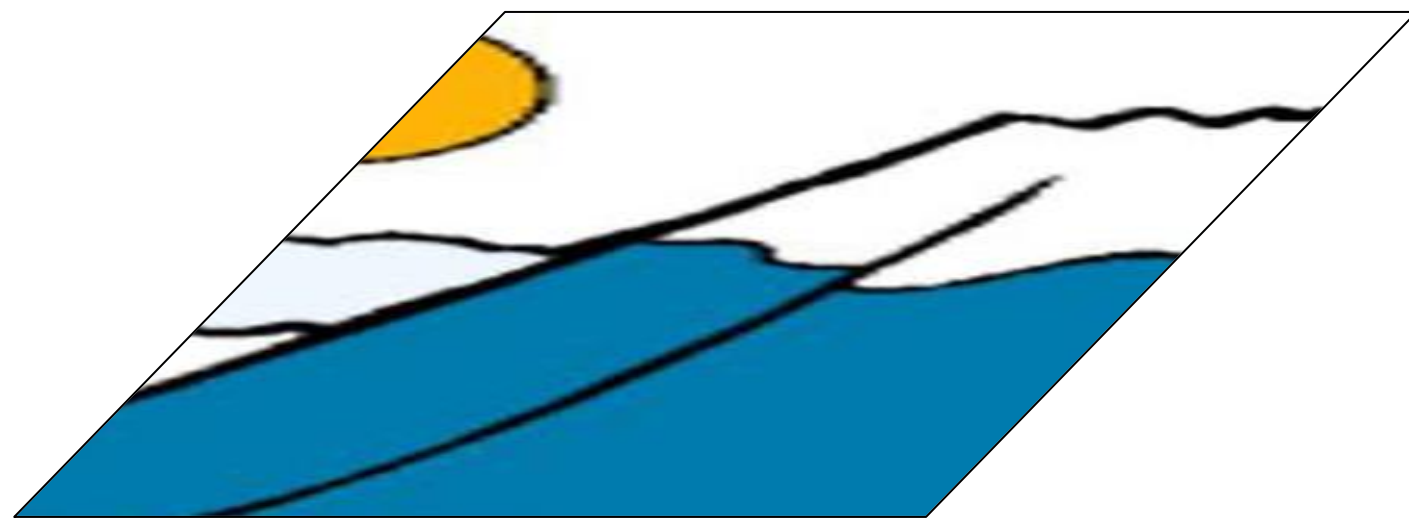


Image 1

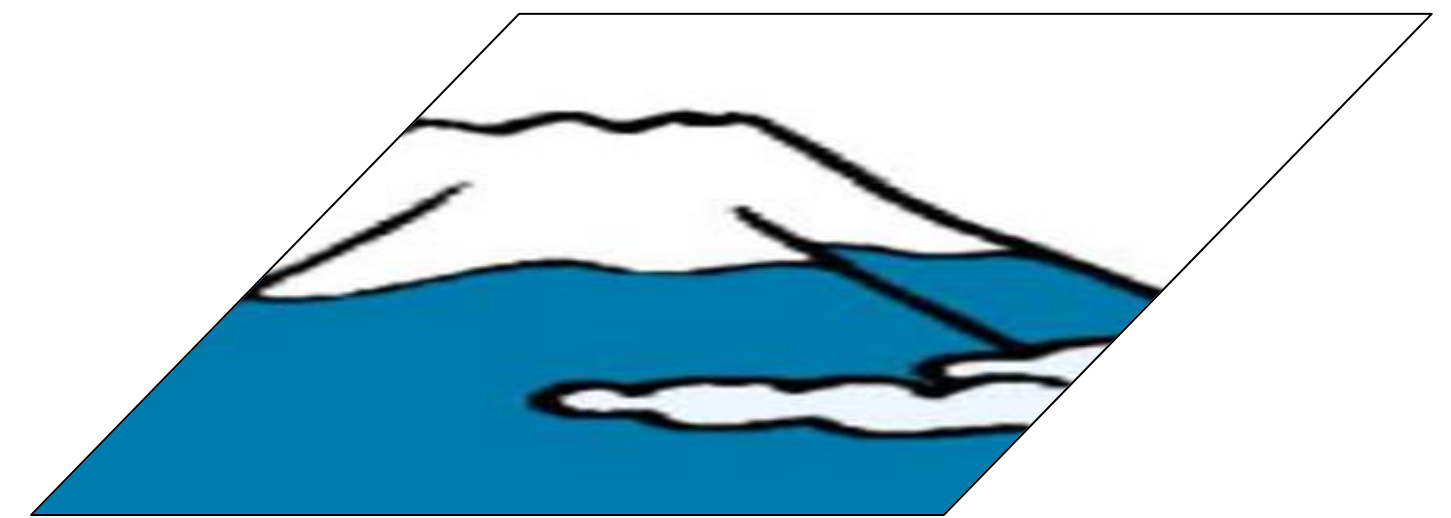


Image 2

Image Pyramids

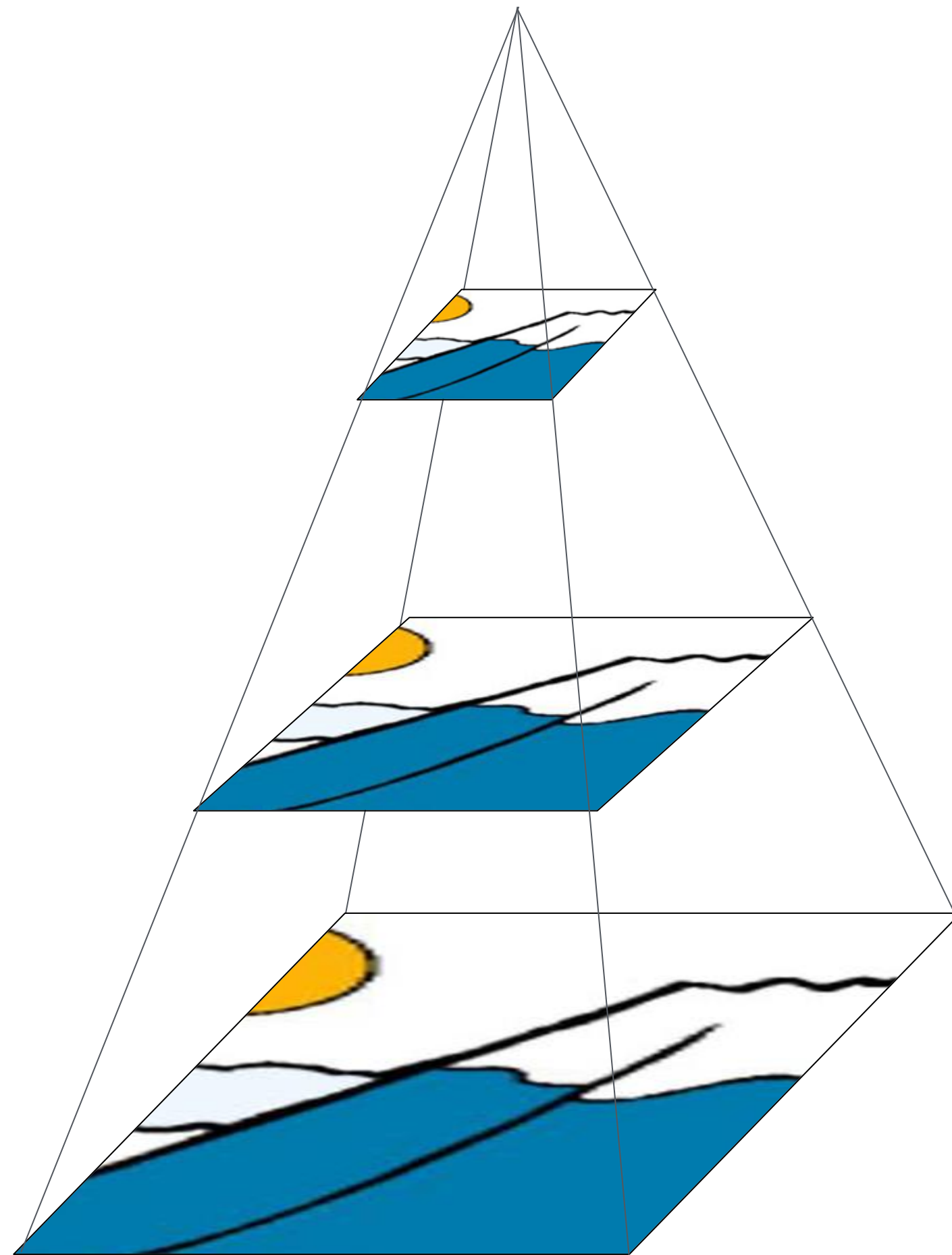


Image 1

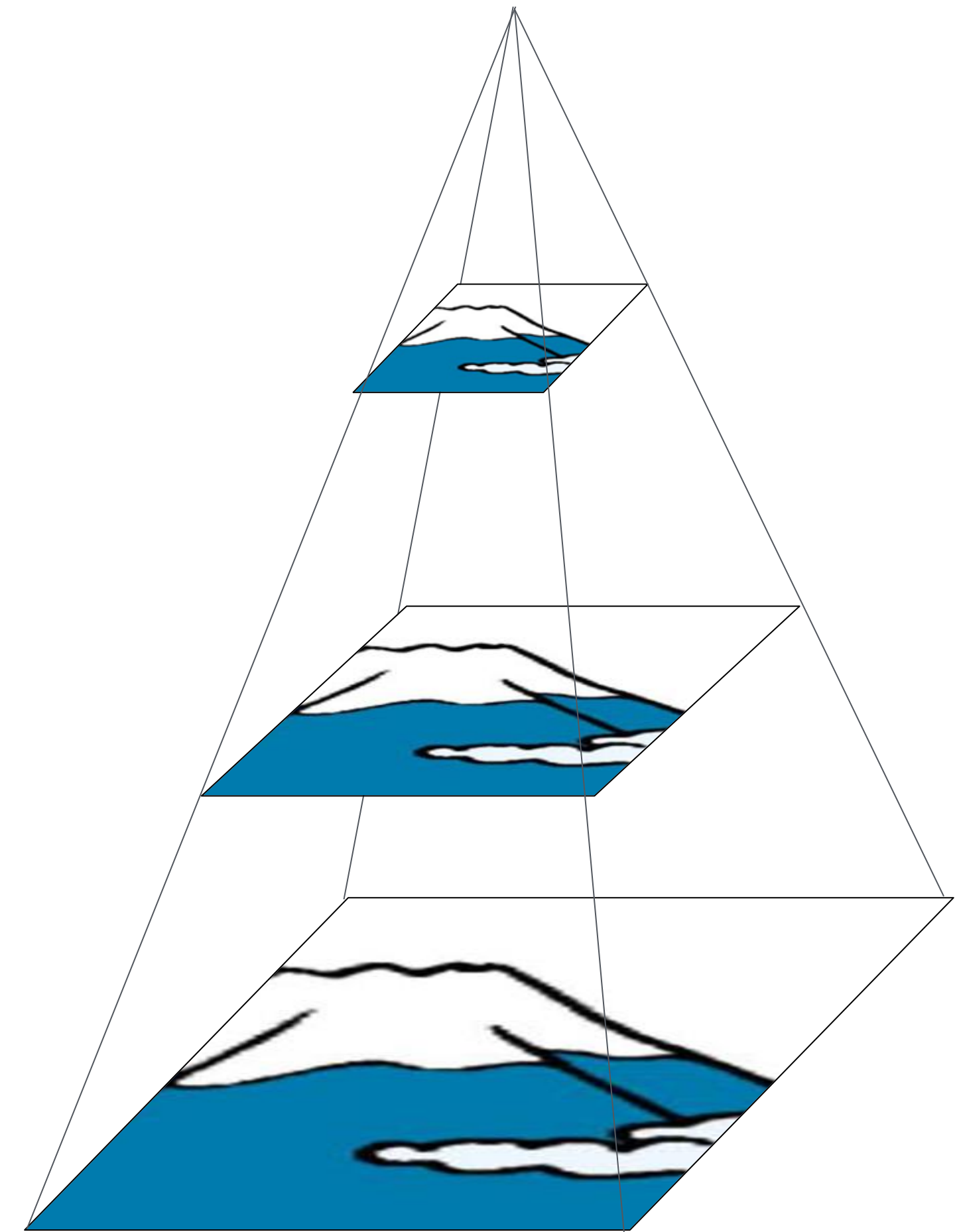
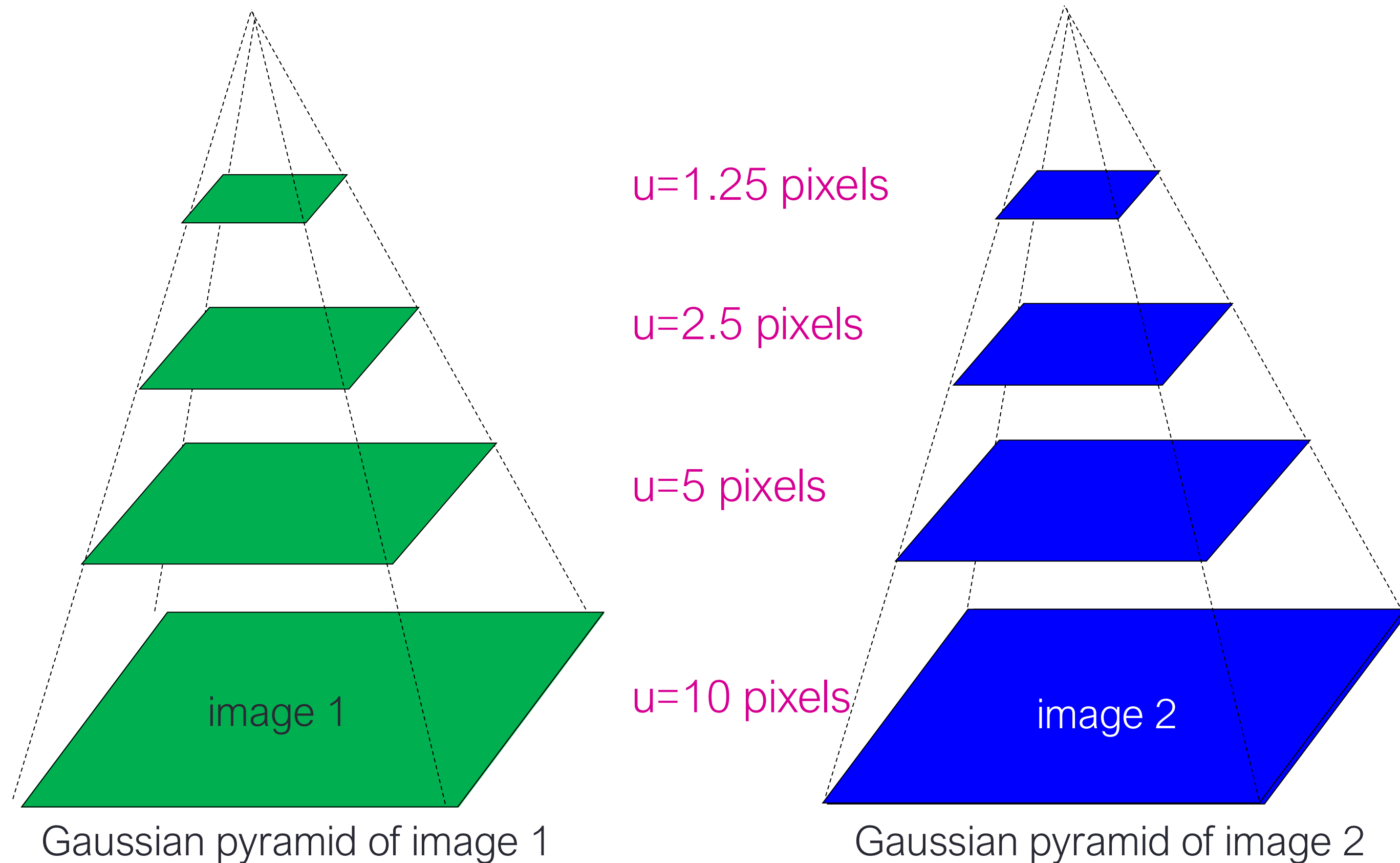


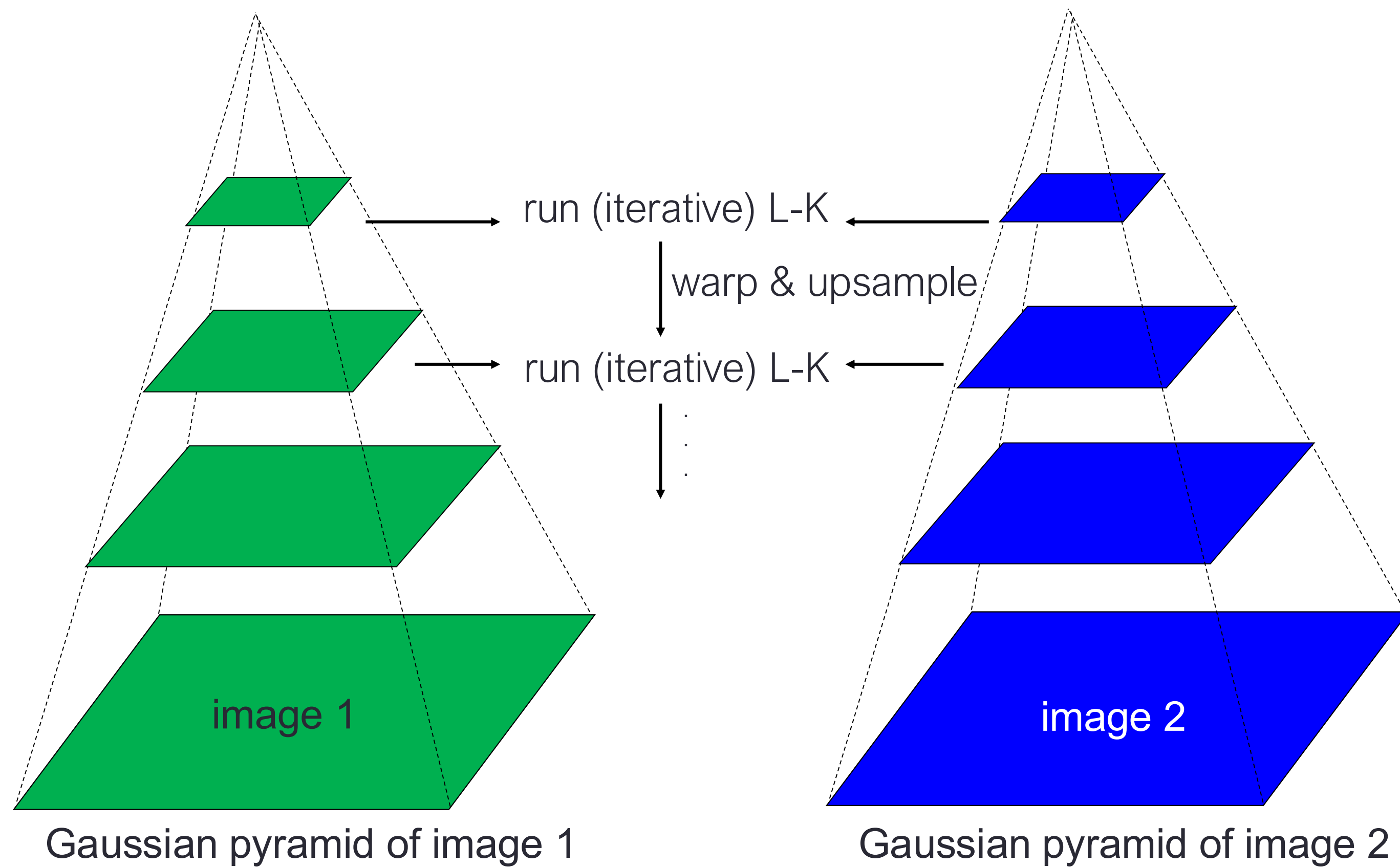
Image 2

Coarse-to-fine optical flow estimation



A coarse-to-fine optical flow estimation, using a Gaussian pyramid, helps solve ambiguities when the "small motion" assumption does not hold

Coarse-to-fine optical flow estimation



Optical Flow

Sparse vs Dense

Compute flow only
for specific features

Compute flow for all
pixels

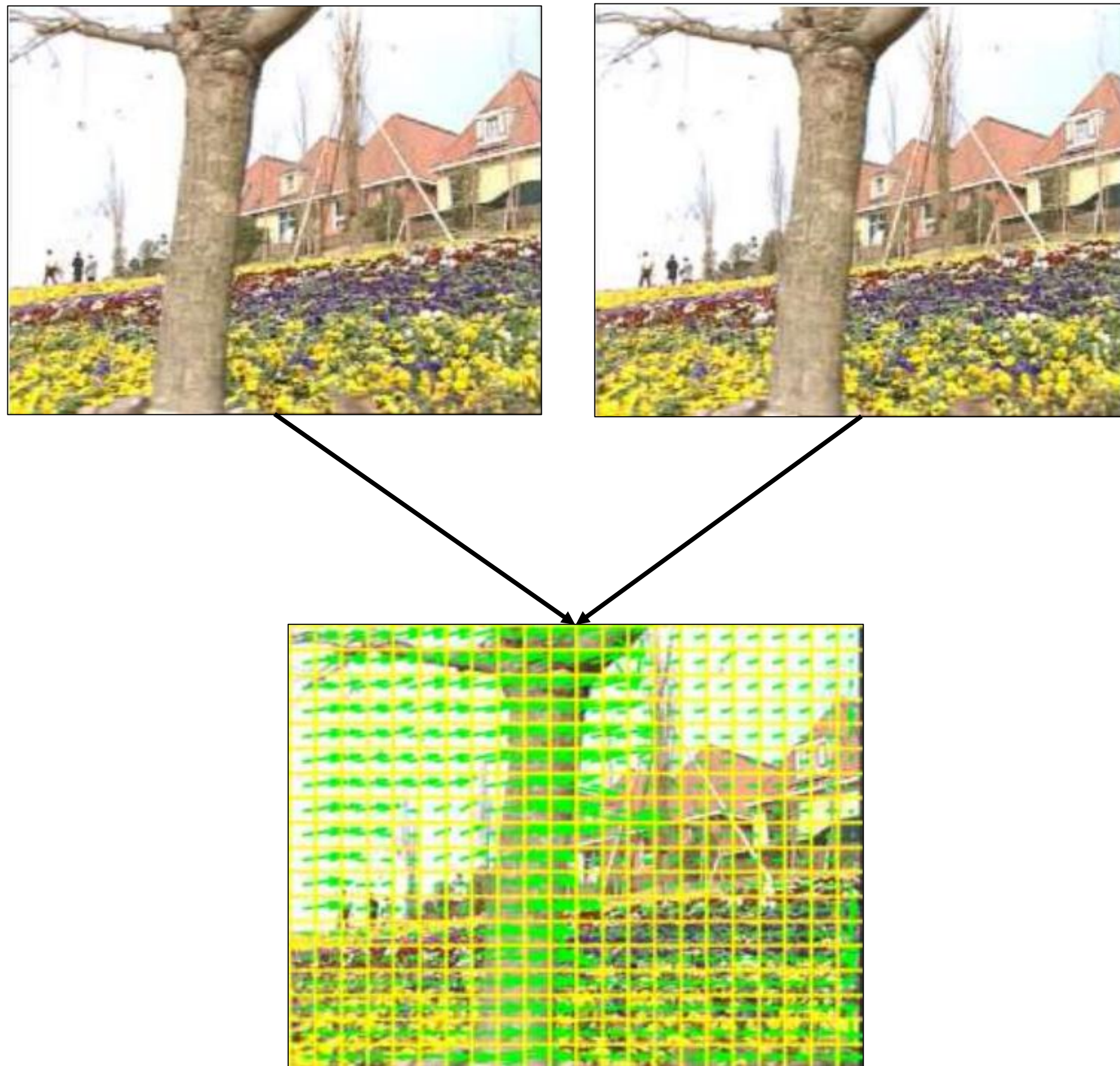
Lucas-Kanade is technically a dense algorithm, but in practice only works on good feature points, i.e., it's sparse.

Farnebäck Optical Flow

- Farnebäck, Gunnar. "Two-frame motion estimation based on polynomial expansion." *Scandinavian conference on Image analysis*. Springer, Berlin, Heidelberg, 2003.
 - Most common approach for dense optical flow estimation
 - Takes second-order Taylor expansion approximation
 - Spatial coherence constraint is not considered (more exact estimation)

Why Optical flow?

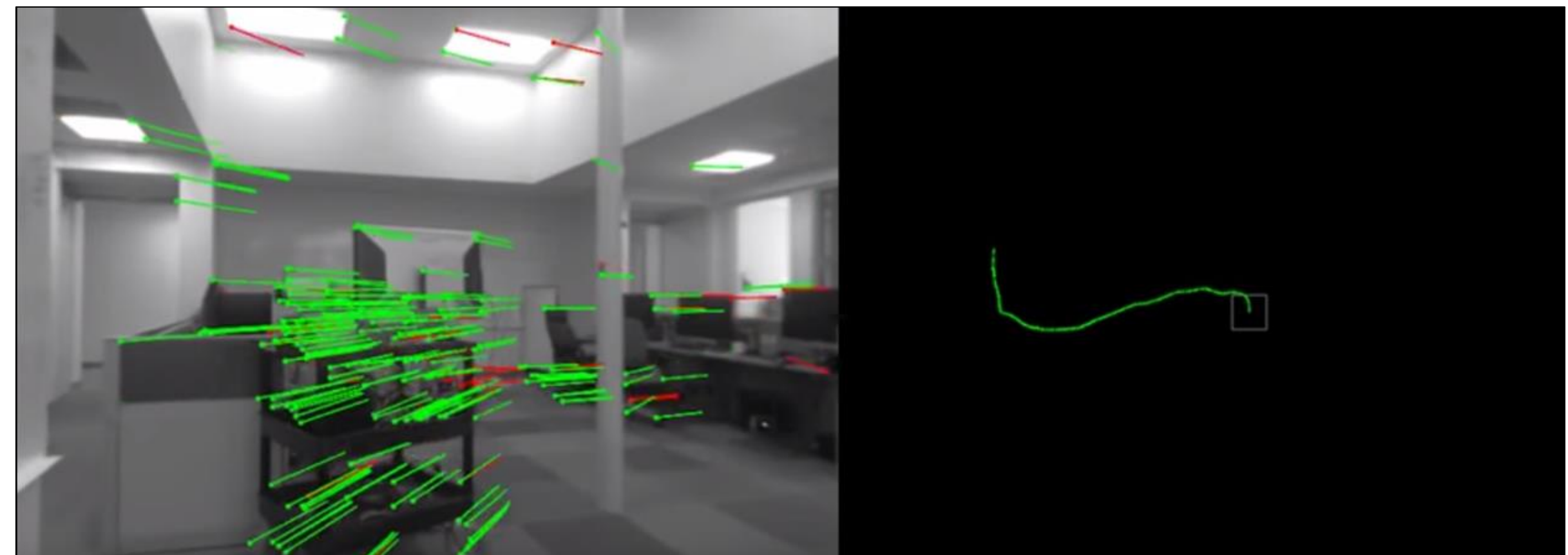
Motion Estimation



Object Tracking



Visual Odometry



Applications of Optical Flow

Autonomous Driving

Estimate ego-motion and detect moving obstacles from vehicle cameras in real time

Action Recognition

Classify human activities in video using flow as a motion feature (two-stream CNNs)

Video Interpolation

Generate intermediate frames for slow-motion, frame rate upconversion, and smooth playback

Video Stabilization

Compensate for camera shake by estimating and removing unintended global motion

Visual Odometry

Recover camera trajectory from frame-to-frame correspondences for robot navigation

Object Tracking

Follow objects across frames using flow vectors to predict position in the next frame

Note: Many of these applications now use deep learning methods (FlowNet, RAFT, PWC-Net) that learn optical flow end-to-end

Many slides adapted from S. Seitz, R. Szeliski, M. Pollefeys, K.
Grauman, J. Hays, among others